



STANDARDS' NON-COMPLIANCE

Product life cycle	Activity	Impact due to
Concept	Usability like ergonomics, man-dated user interface like avionics	Ergonomics—Product usage doesn't impact the user. Complying to standards laid out by the designated regulatory agencies like FAA/FDA etc.
Design	Products need to meet safety, emission, and environment requirements	- Compliance to safety is a design issue which also increases the design time and cost. Incorporating them early is a must as at the end of design compliance issues ultimately end up in redesign, leading to an escalation in cost and time to market. - The situation is the same for products which need to meet stringent environmental standards compliance by way of selecting the right components, right manufacturing process, and low energy consumption; it calls for design of the product factoring these needs.
Engineering (NPI)	Component selection, material selection for enclosures, plastic/die-cast tooling and ability to recycle	- Most important aspect is the component selection. Right component selection is key to the manufacturing process as well as products complying to standards. Good example is industrial-grade components, which withstand high temperature and vibration compared to commercial version. Right components are also needed for complying to green manufacturing processes. - Choice of materials for the enclosures. It could be sheet metal, metal diecast, or plastic, depending on the environment the products work in and corresponding standard applicable. - Third key is the ability to recycle the product after its active life. This calls for first the right design and second proper material selection.
Manufacturing	In electronics manufacturing, the main compliance today is that manufacturing lines have to comply to the avoidance of lead in the process and any traces of usage of CFC (which earlier processes used)	- Today, almost all products have to comply to the RoHS (Reduction of Hazardous Substance – aka lead-free) process. In addition, they have to document the composition of the materials used in the products as per the REACH standard. - Another aspect that is missed is the packing materials used for shipping. Most European countries mandate these have to be compliant to green laws, especially to be pest-free and safe to decompose.
Support	This phase concerns repair of faulty products. More critical is the management of component obsolescence. In addition, making the products comply to standards	- Impact of standards is in two ways. One, when the components become obsolete, the replacement normally calls for a redesign as very rarely you get a drop-in compatible part. This calls for retesting, which is time consuming and needs planning. - Second, when the product is repaired, replication of the process to meet the standard is critical. For example, most industrial products are coated with a 'conformal coating' to prevent moisture impacting the product. Normally, the coating is done when all the testing and calibration of the product/PCB is complete. So, when the PCBA is repaired, the technician removes the coating to repair but most of the time does not reapply the coating, which leads to moisture impacting.
Withdraw	Finally, when the product is discarded, most countries mandate it is safely dismantled and recycled, or disposed of (disposable part is much less than 1%)	Unless the design addresses the issue early on, meeting the recycling standards will be a challenge. And if not incorporated, practically it is very difficult to do it later. This again shows that the importance of compliance to standards has to be addressed early on.

3. Safe electrical emission and susceptibility

Complying to the standards is mandatory and exporting to different countries depends on the standards laid. Typically, standards are defined by international agencies like FDA, FAA, IEEE, IEC, etc. In India we have Bureau of Indian Standards (BIS) (earlier was known as Indian Standards Institute) which has standards for products made and used in India. Almost all the countries have made compliance to standards mandatory.

For complex products like aircrafts and military systems, most countries tend to follow FAA (Federal Aviation Administration) and FDA

(Food and Drug Administration) standards. Similarly, standards like Electrical Radiation Emission and Susceptibility, FCC (Federal Communication Commission) of the USA, or CE (Conformité Européenne) for European Union are used across countries as these standards have been well drafted and have stood the test of time. With the increased integration into global economy, India is also now mandating standards for use in India.

Most designers tend to assume that compliance to standards is a separate step and can be done once the design is complete. However, compliance to standards needs atten-

tion from the design phase. In fact, one of the design parameters should be the standards to which the product has to comply. For example, if the product has to meet FCC Class B emission level, it is better to have metallic enclosure rather than a plastic enclosure—if the design allows. So, designers need to understand the standards to which the product should comply and ensure the design goals meet that.

Why this is critical

First and foremost reason for compliance is the impact on business. Products that do not meet the standards cannot be sold in the market.